Whistlegraph

Drawing, Singing, and the Graphic Score as Viral Form

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1. Introduction

"I'm a butterfly. Flapping for you guys. It's just a costume. I put on in my room." A hand draws a butterfly on paper while a voice sings these words. Each wing is a line; each line is a lyric; the finished drawing is a score that anyone can perform by tracing the lines and singing the words. The video is 15 seconds long. It was posted on TikTok in early 2020. Millions of people watched it.

This is a whistlegraph: a memorizable hybrid of drawing, singing, and poetry in which every mark corresponds to a sung syllable, producing a graphic score that is simultaneously artwork, song, and instruction manual for its own reproduction [Dirt Magazine, 2023]. I invented it in 2019, developed it into a trio practice with Alex Freundlich and Camille Klein during the COVID-19 pandemic, scaled it to 2.6 million TikTok followers, exhibited it at the New Museum and on Feral File, and then—after the trio separated in November 2023—it fed directly into the design of Aesthetic.Computer.

This paper is not a retrospective. It is documen-

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tation of a form that emerged from the intersection of several practices—Radical Digital Painting [@jeffrey, 2017], Goodiepal's Radical Computer Music [Goodiepal (Gaeoudji SYGNOK), 2012], constraint-based art, and short-form social video—and that continues to inform my work. The whistlegraph is the bridge between my painting practice and the creative computing platform.

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2. The Form

2.1 Structure

A whistlegraph has three simultaneous layers:

1. **Visual**: A drawing made in real time, typically with pen or marker on paper, or chalk on blackboard. Each stroke is deliberate and corresponds to a musical element.
2. **Musical**: A sung melody. Each note aligns with a drawn stroke. The melody is simple enough to memorize after one or two viewings.
3. **Textual**: Lyrics that function as poetry. The words describe, narrate, or instruct. They are short enough to remember.

The finished drawing serves as a *graphic score*—a visual document that encodes all the information needed to reproduce the performance. Anyone who can see the drawing and recall the melody can perform the whistlegraph. The score teaches you how to play it.

This self-documenting property distinguishes the whistlegraph from other graphic scores in experimental music [Magnusson, 2010]. Cage's or

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Figure 1: “I’m a Butterfly (Cosplayer)”—the graphic score that launched the viral cycle. Four strokes, four sung lines, one memorizable image.

Cardew’s graphic scores require interpretation—a performer must decide what the visual marks mean musically. A whistlegraph requires no interpretation. The relationship between mark and sound is one-to-one, established during the original performance and preserved in the recording. The score is not a prompt for improvisation; it is a set of precise instructions disguised as a picture.

2.2 Duration and Constraint

Most whistlegraphs last 5–30 seconds. The constraint is deliberate: the form must be short enough to memorize in one viewing, simple enough that a child can reproduce it, and complete enough that the finished drawing is satisfying as a standalone image. These constraints—brevity, simplicity, completeness—are the whistlegraph’s design parameters. They

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echo Klee's pedagogical exercises [Klee, 1925] and Kandinsky's point-line-plane analysis [Kandinsky, 1926], but translated into a format native to TikTok's 15-second vertical video.

The longest whistlegraph—"The Longest Whistlegraph Ever (so far)," commissioned by Rhizome [Rhizome, 2022]—lasted approximately 22 minutes and was performed in chalk on a 4×6-foot blackboard with a bell on top. It was the exception that proved the constraint: the audience at the New Museum watched in sustained attention because they had been trained, through hundreds of 15-second whistlegraphs, to expect the form's characteristic fusion of mark and melody.

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3. Origins

3.1 Radical Digital Painting

The whistlegraph emerged from my Radical Digital Painting (RDP) practice, active since 2016 [@jeffrey, 2017]. RDP treated digital painting not as a supplement to traditional painting but as its next material ground: "Painting is just a kind of picture message." Over 65 lecture-performances across the US and Europe—including a 2018 European tour with Goodiepal & Pals and a presentation at the 35th Chaos Communication Congress [@jeffrey, 2018]—developed a format in which painting, speaking, and performing were fused into a single act.

The whistlegraph formalized what the RDP lectures did informally: synchronized the act of making with the act of narrating. Where an RDP lecture was improvised and unrepeatable, a whistlegraph was composed, memorizable, and reproducible. The shift from improvisation to composition—from lecture to score—was the key innovation.

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3.2 COVID-19 and the Trio

In 2020, during the COVID-19 lockdown, I formed a trio with Alex Freundlich and Camille Klein in a cabin in Ashland, Oregon. The pandemic eliminated the possibility of live performance and lecture tours. TikTok, which had exploded in popularity during lockdown, offered a platform for short-form video that matched the whistlegraph's natural duration.

The trio posted musical drawing tutorials: a hand draws while a voice sings, the camera captures both, and the result is a 15-second video that teaches the viewer to reproduce the whistlegraph themselves. The format was natively suited to TikTok's duet and stitch features, which allowed viewers to perform alongside the original video. The whistlegraph was not just content to be consumed; it was a score to be performed.

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4. Virality

4.1 Growth

The @whistlegraph TikTok account grew from zero to 2.6 million followers between 2020 and 2023. The growth was organic—no paid promotion, no algorithmic gaming, no trend-jacking. The viral mechanism was the form itself: viewers who learned a whistlegraph wanted to perform it, and their performances introduced the form to their own audiences.

"I'm a Butterfly"—a collaboration with musician bo en—was the earliest viral hit. Its melody ("I'm a butterfly / flapping for you guys / it's just a costume / I put on in my room") and its drawing (a simple butterfly rendered in four strokes) were memorable enough to spread through repetition rather than recommendation.

4.2 Why It Worked on TikTok

TikTok's dominant content formats—dance challenges, lip-sync, comedy sketches—share a structural property with the whistlegraph: they are *re-*

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producible performances. A dance challenge succeeds when viewers learn the choreography and post their own version. The whistlegraph is a drawing challenge with the same viral mechanics: learn the score, perform it, post it.

But the whistlegraph introduced an element absent from other TikTok formats: *original composition.* Dance challenges use existing songs. Lip-sync uses existing audio. The whistlegraph is its own song, its own drawing, and its own instruction manual. It does not borrow virality from another medium; it generates it from its own structure.

This is the Papert connection [Papert, 1980]: the whistlegraph is a "microworld" in which the learner constructs knowledge by building something. The viewer does not passively consume the whistlegraph; they actively reproduce it, and in doing so, learn something about the relationship between visual mark and musical note.

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5. Institutional Reception

Institution	Work	Year
Rhizome / New Museum	Commission + performance	2022
Feral File	Ten Whistlegraphs (45 ed.)	2022
Taipei Dangdai	NFT Launch Stage	2022
Schneider Museum (SOU)	Outdoor installation	2021
KADIST (San Francisco)	RDP #98 in collection	2021
SMK (Denmark)	Ten Minute Painting	2018
Rhizome ArtBase	Permanent accession	2022

Table 1: Institutional exhibitions and collections.

The institutional trajectory is notable for its speed: from TikTok tutorials to a Rhizome/New Museum commission in under two years. The Rhizome commission [Rhizome, 2022]—part of the First Look series co-presented with the New Museum—included a live chalk performance, a 22-minute 4K video, token-gated manuscripts and recordings on IPFS, and workshops for ages 13–18 in Queens (supported by Deutsche Bank

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Figure 2: The ten graphic scores from the Feral File exhibition *Ten Whistlegraphs* (2022). Each score encodes a complete performance: drawing, melody, and lyrics. Top row, left to right: Butterfly Cosplayer, Time to Grow, I Don't Need an iPhone, Lately When I Fly, Loner. Bottom row: Mommy Wow, People Pleaser, Slinky Dog, Puzzle, What's Inside Your Heart.

Americas Foundation). The work was accessioned into Rhizome's ArtBase, placing a TikTok-native art form in the same collection as net art by Olia Lialina and JODI.

The Feral File exhibition [Whistlegraph, 2022]—ten whistlegraphs as ERC-721 tokens, each including video performance, graphic score, and curated TikTok selections—sold 45 editions.

It was also shown at Taipei Dangdai alongside Art Blocks and Outland.

A 60-page zine, edited by Asher Penn (*Sex Magazine*), was published in an edition of 750 copies in

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June 2023. It included interviews with all three trio members and an essay by Jacob Ciocci (Paper Rad) titled “The Butterfly Effect / Rules Set You Free” [Ciocci, 2023], connecting the whistlegraph to Paper Rad’s legacy of rule-based, community-generated art.

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6. The Music

The trio released music on streaming platforms: singles (“Infinitely Falling, Pop Sounds,” “Whistlegraph Practice,” “What’s Inside Your Heart?”) and a studio album, *Music 2 Whistlegraph 2* (December 2022, produced by Charlie Kamin-Allen). The album extended the whistlegraph form into longer compositions while retaining the core principle: every musical element corresponds to a visual mark.

The music was not a side project. It was the whistlegraph’s other half. The graphic score without the music is a drawing; the music without the graphic score is a song; together, they are a whistlegraph. The streaming releases documented this fusion in a format that could circulate independently of TikTok.

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7. Separation and Continuation

In November 2023, the trio separated due to a conflict in future goals. I took sole access to the group's social media platforms, including the 2.6-million-follower TikTok account. The @whistlegraph handle now posts Aesthetic.Computer content.

The separation ended the trio but not the form. The whistlegraph's core properties—reproducibility, mark-melody correspondence, the graphic score as instruction manual—directly informed the design of Aesthetic.Computer:

- The **prompt-as-instrument** metaphor (memorizable paths discovered through improvisation) echoes the whistlegraph's memorizable scores.
- The **piece model** (self-contained programs that run in one context) echoes the whistlegraph's self-contained performances.
- The **immediate-mode renderer** (every frame drawn from scratch) echoes the whistlegraph's insistence that the drawing happens in real time, not as a static artifact.

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- The **TikTok-to-platform pipeline**—2.6 million followers on a social network feeding into a creative computing platform—is a distribution model that no other creative tool has attempted.

The Whistlegraph was not a detour from Aesthetic.Computer. It was the audience-building phase that most creative computing tools never have. Processing had MIT; p5.js had UCLA; Sonic Pi had Cambridge. Aesthetic.Computer had 2.6 million people who learned that drawing and music could be the same act.

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8. The Graphic Score as Interface

The whistlegraph proposes that a drawing can be an interface—not a representation of an interface (like a wireframe) but a *functional* interface that encodes instructions for its own execution. This is the connection to Aesthetic.Computer's design philosophy: the prompt is an interface that teaches you how to use it by responding to your input. The piece is an interface that reveals its affordances through interaction. The whistlegraph is an interface that reveals its melody through tracing.

Eshun's "sonic fiction" [Eshun, 1998]—music that constructs the reality it describes—has a visual analogue in the whistlegraph: a drawing that constructs the performance it depicts. The graphic score is not a document *about* the whistlegraph; it *is* the whistlegraph, in the same way that a piece of code is not a document about a program but the program itself.

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9. Conclusion

The whistlegraph is a form that should not have worked on TikTok. It is not a dance. It is not a lip-sync. It is not a comedy sketch. It is a person drawing slowly while singing quietly. It reached 2.6 million followers because it offered something TikTok's native formats do not: a score you can learn, a drawing you can make, a song you can sing, and instructions for doing all three simultaneously.

The form's institutional reception—Rhizome, New Museum, Feral File, KADIST, SMK—confirms that the whistlegraph is recognized as art. Its TikTok reception confirms that it is recognized as content. Its continuation in Aesthetic.Computer confirms that it is recognized as a design methodology. It is all three, because it was designed to be reproducible, and reproducibility is the property that connects art (the score), content (the video), and interface design (the prompt).

The author "sacrificed the joy of endlessly drawing with ease for the pain of advancing the tools

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of his trade" [@jeffrey, 2019]. The whistlegraph was the last project in which drawing was easy. Everything since has been building tools.

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